



SiteManager

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Introduction

Value Proposition

“Right people. Right equipment. Right on time.”

Problem

Construction companies require a digital dashboard to track all projects, employees, and equipment in one centralized location, as current systems often lack real-time visibility, resulting in inefficiencies. Managers frequently have to rely on manual reports, calls, or spreadsheets to understand what resources are available or where progress is being delayed, which slows down operations and leads to scheduling conflicts.

Solution

Our solution is to create a live map that links to projects and shows exactly which employees, equipment, and materials are needed at each location. By combining this information into one dashboard, SiteManager helps construction managers make faster, data-driven decisions, reduce downtime, and improve communication between teams on different sites. The result is a smoother, more efficient workflow that keeps projects on track and on schedule.

Prototype

Figure 1: Webpage Dashboard

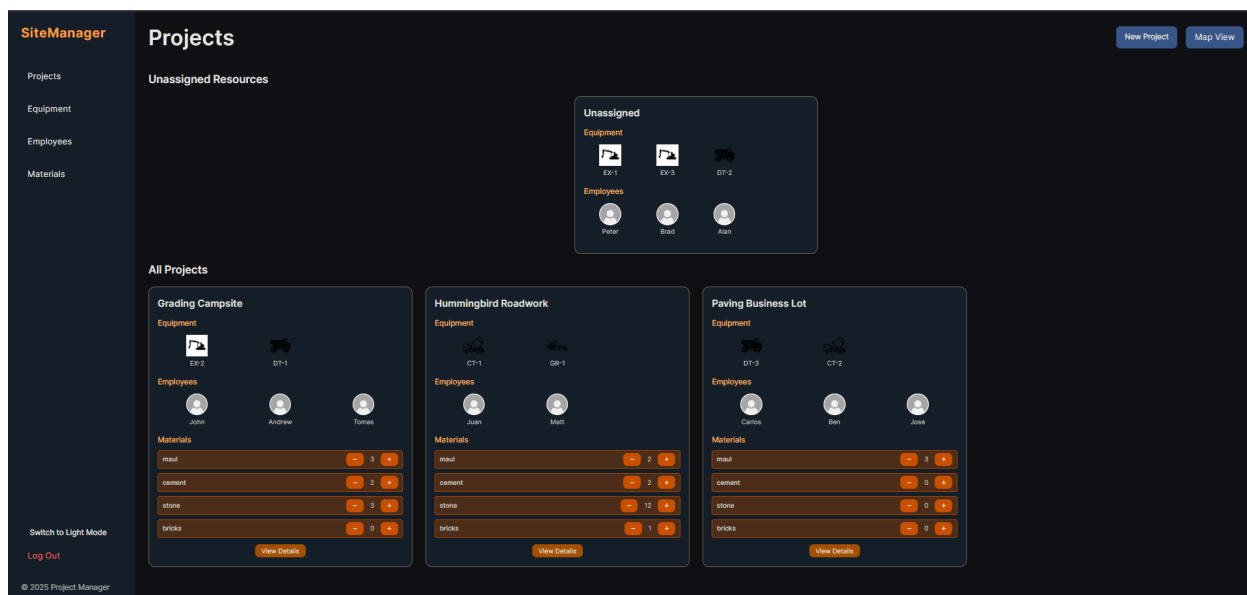
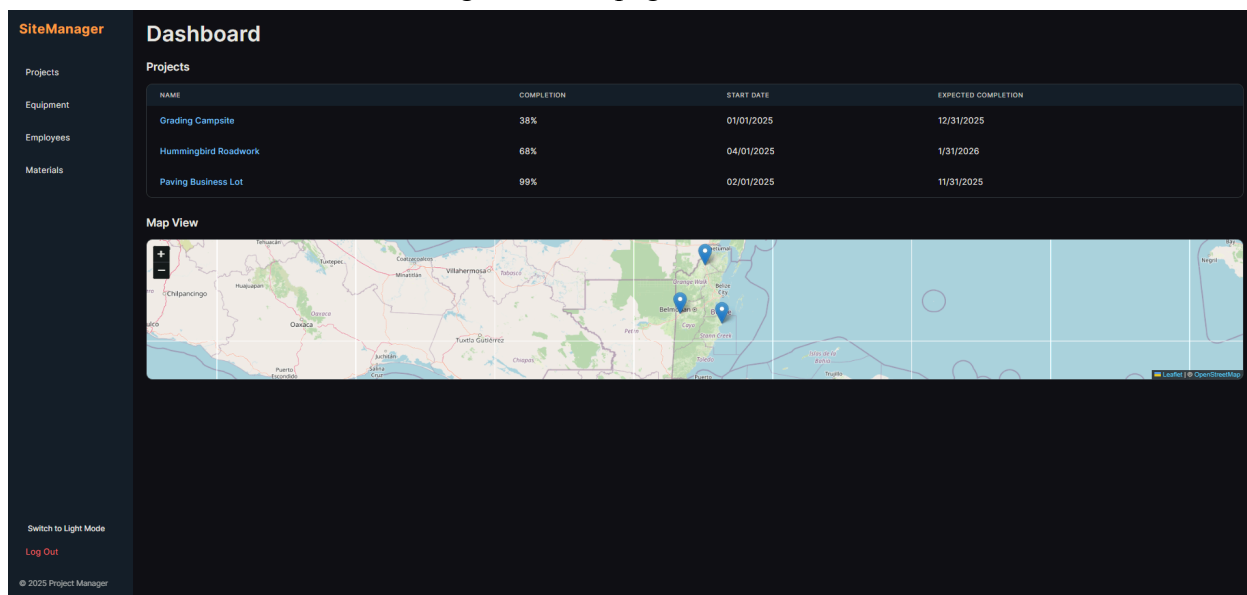


Figure 2: Webpage Projects

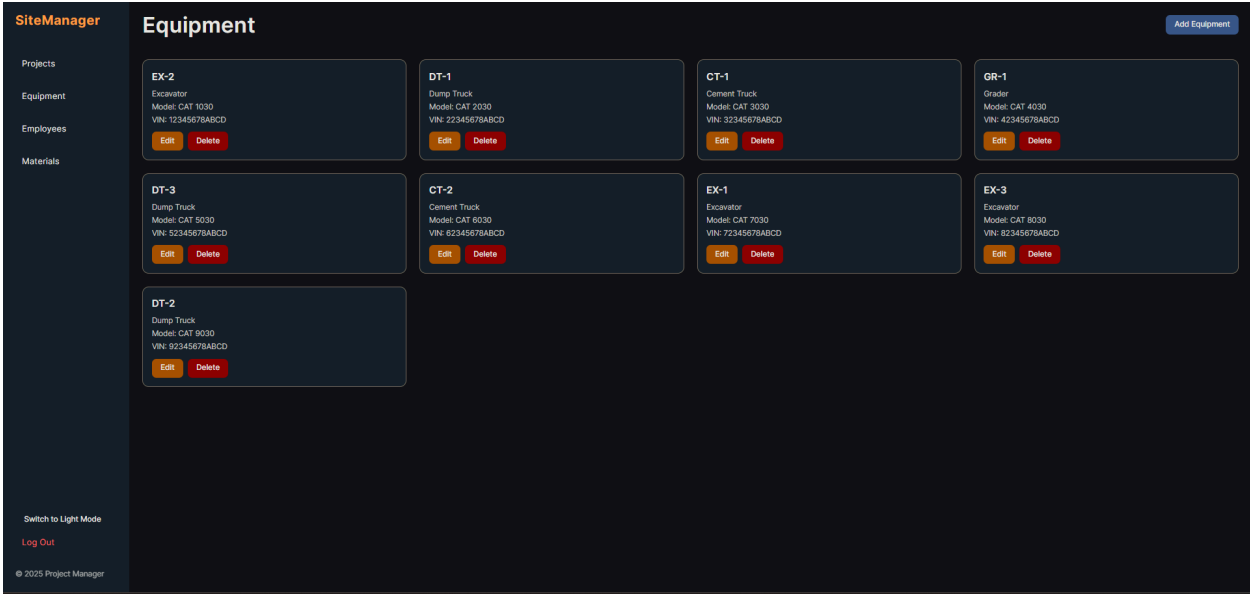


Figure 3: Webpage Equipment

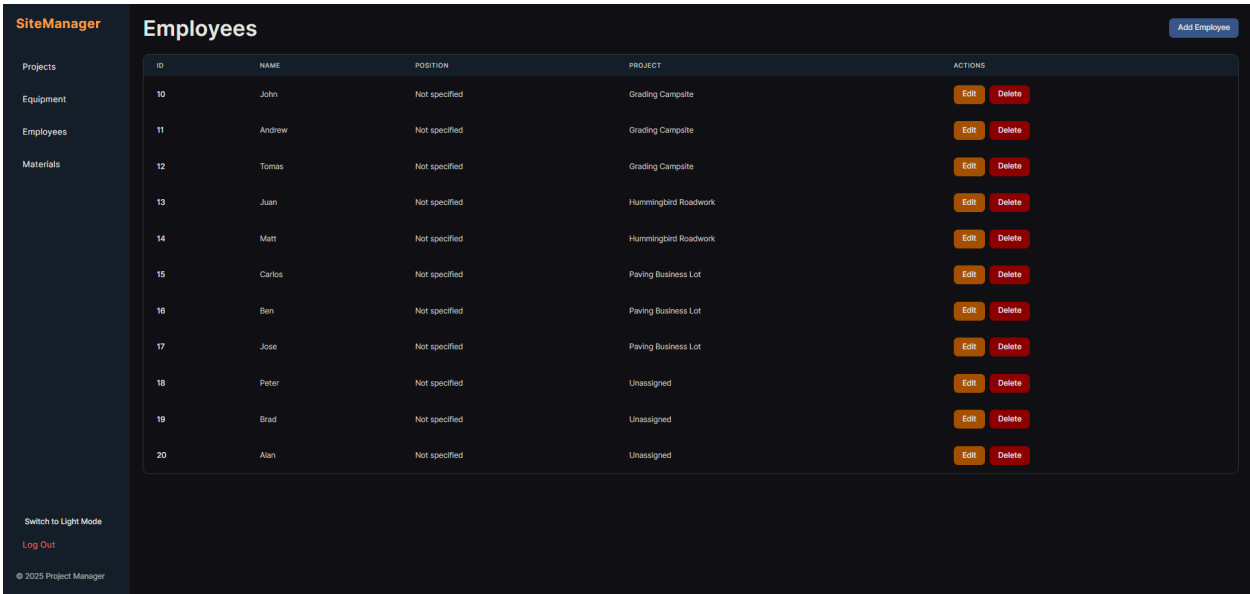


Figure 4: Webpage Employees

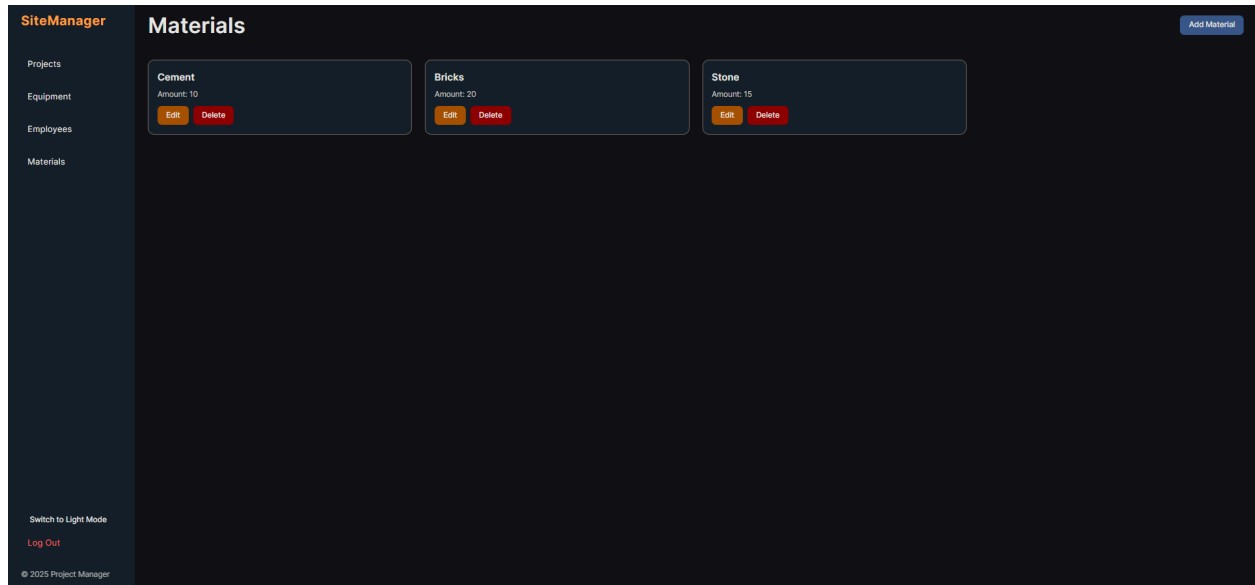


Figure 5: Webpage Materials

Tasks & Final Interface Scenarios

Web:

Task 1 (Easy): Store the list of heavy equipment and their maintenance records and status, allowing the mechanic to edit as required.

Task 2 (Medium): Store the list of all ongoing projects and allow the foreman to see required equipment/materials and upload progress notes.

Task 3 (Difficult/Complex): Show all ongoing projects on an interactive map. Each project contains a list of assigned employees, equipment, and materials that can be edited by the manager.

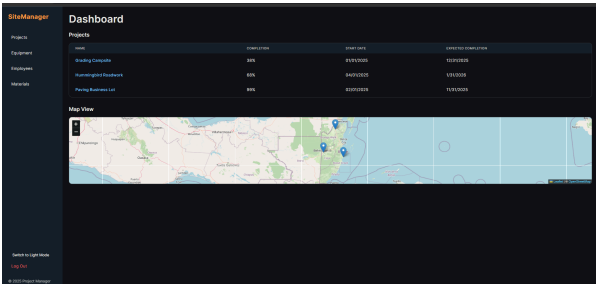
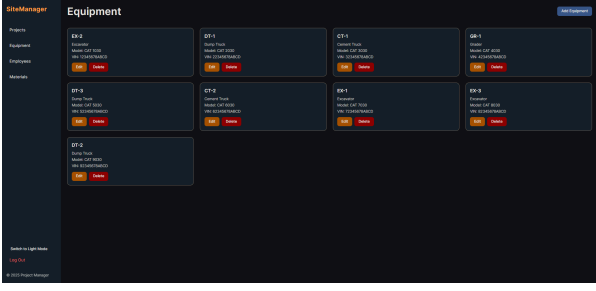
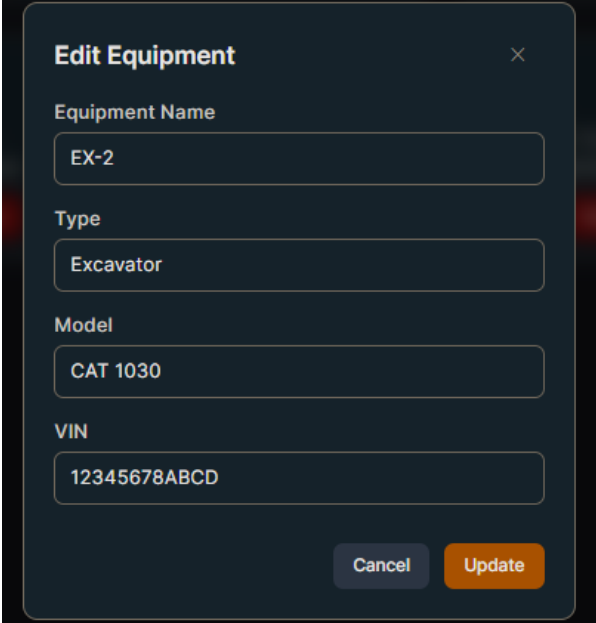
Why we chose Task 1: Tests fundamental data entry and simple status visualization.

Why we chose Task 2: Tests the application's ability to link multiple data types onto a single screen.

Why we chose Task 3: Tests the integration of simultaneous multi-entity editing and ensures data consistency across all views.

Storyboard for 3 tasks

1. Task 1: Store the list of heavy equipment and their maintenance records and status, allowing the mechanic to edit as required.

	The user opens the application, and the Site Manager Dashboard is displayed.
	The user clicks “Equipment” in the sidebar to view the equipment list.
	The User Locates the ‘CAT 1030’ Excavator and clicks the “Edit” button. The user proceeds to change the text in the “VIN” field,

Edit Equipment

✕

Equipment Name

Type

Model

VIN

Cancel

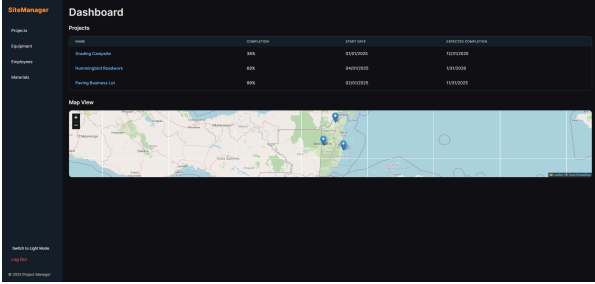
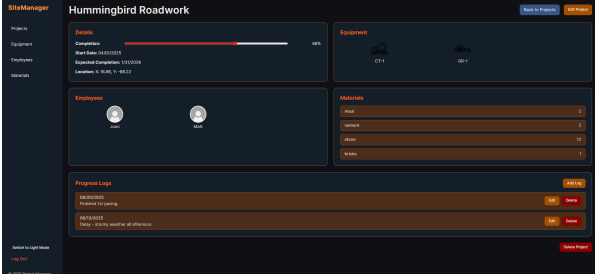
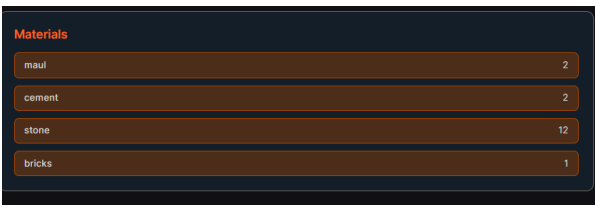
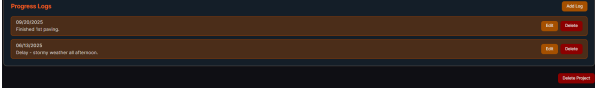
Update

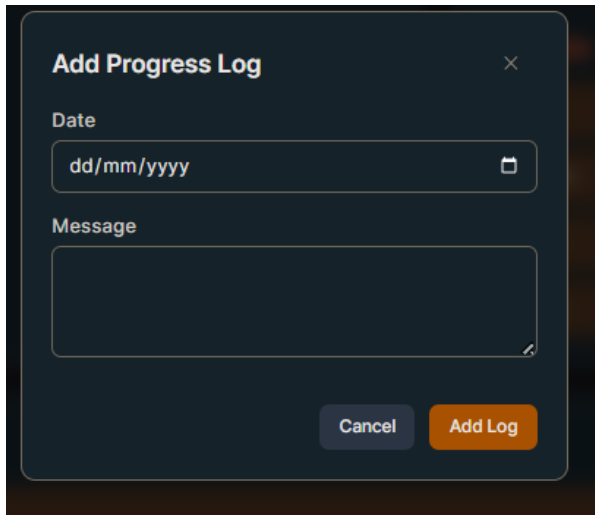
The user clicks “Update” to save the changes,

[illegible]

The Equipment displays the edited version of 'CAT1030' Excavator.

2. Task 2: Store the list of all ongoing projects and allow the foreman to see required equipment/materials and upload progress notes.

	The user accesses the App's Dashboard to find project status.
	The user clicks on the “Projects” on the sidebar to view project overview.
	The user clicks “Hummingbird Roadwork” and views the project’s details.
	The user views the ‘Materials’ section to check on the remaining amounts of supplies.
	The user scrolls down the page to the “Process Logs” section and clicks “Add Log.”



Add Progress Log ×

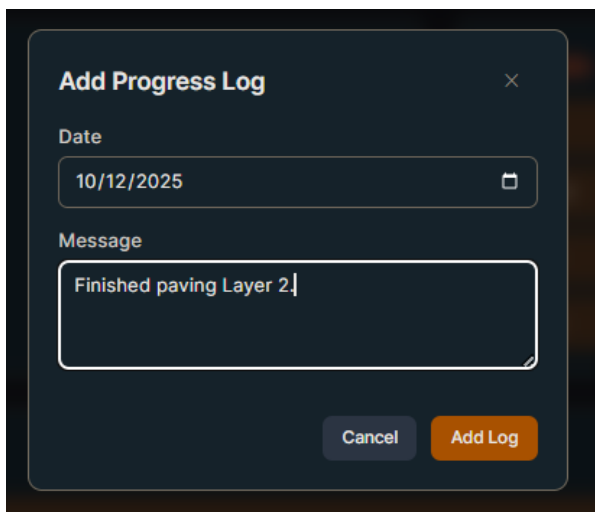
Date

dd/mm/yyyy

Message

Cancel Add Log

The user clicks on “Add Log.”



Add Progress Log ×

Date

10/12/2025

Message

Finished paving Layer 2

Cancel Add Log

The user selects the date and enters new progress information, then clicks “save.”

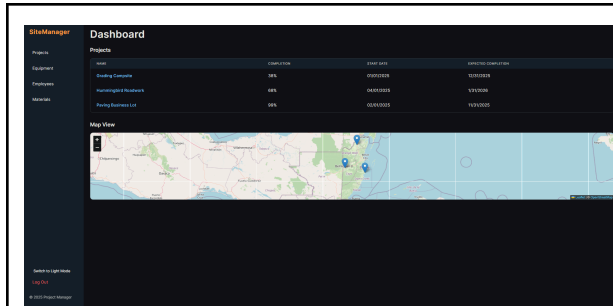


Progress Logs		Add Log
10/12/2025	Finished paving Layer 2	Edit Delete
09/12/2025	Finished to paving	Edit Delete
09/12/2025	Delay - stormy weather all afternoon.	Edit Delete
09/12/2025	Finished paving Layer 2	Edit Delete

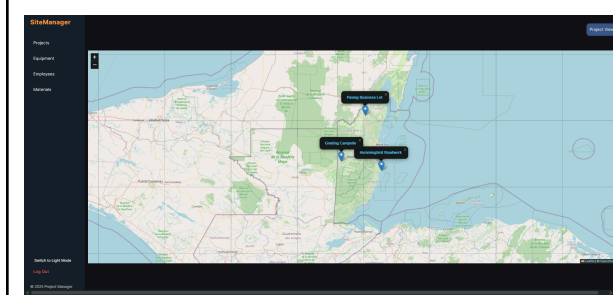
Dismiss Project

The new log appears at the top of the “Progress Logs” list with the date.

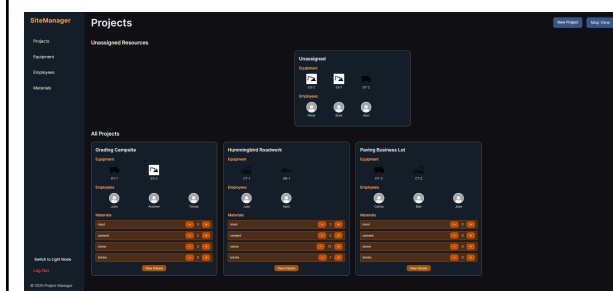
3. Task 3: Show all ongoing projects on an interactive map. Each project contains a list of assigned employees, equipment, and materials that can be edited by the manager.



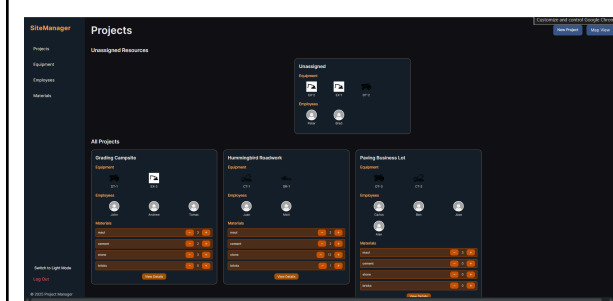
The user accesses the App's Dashboard to find the "Map View."



The user clicks on the "Map View" and searches for 'Paving Business Lot,' then clicks on it.



The user then prompts to open the Projects tab via the pin in the previous step.



The user drags and drops 'Alan' over to the 'Paving Business Lot.' The system updates the 'Paving Business Lot' card, showing 'Alan' now assigned.

ID	Name	Role	Project	Actions
01	John	Not specified	Marketing Strategy	Edit Delete
02	Jane	Not specified	Marketing Strategy	Edit Delete
03	Mike	Not specified	Marketing Strategy	Edit Delete
04	Sarah	Not specified	Product Development	Edit Delete
05	David	Not specified	Product Development	Edit Delete
06	Emily	Not specified	Product Development	Edit Delete
07	Chris	Not specified	Product Development	Edit Delete
08	Alex	Not specified	Customer Support	Edit Delete
09	Olivia	Not specified	Customer Support	Edit Delete
10	Noah	Not specified	Customer Support	Edit Delete

Search & Filter Status
Log Out
© 2024 Project Manager

The user navigates to the “Employees” list via the sidebar. The user verifies that Alan is now listed for the specific project. (In this screenshot, Alan is not shown as assigned because we do not have a backend as yet)

Design Evolution

Sketches

Home Projects Map Equipment Employees Logout

New

Grading Complete

Star Roadwork

Star Parking Road

Unassigned

Employees V

Equipment V

Materials V

Expand Collapse

Write Mail Comment

Home Projects Map Equipment Employees Logout

Equipment → DT1

Orange Walk

Grading Complete

Aug 14, 2025

S: DTR-108EN

ABCDEF2259

Maintenance Log

Issue	Date	Urgency
Oil Leak	08/01/25	Minor
Broken AC	08/01/24	Fixed
Transmission	03/01/24	Severe

Edit Inactivate Delete

Home Projects Map Equipment Employees Logout

Name	Position	Current Project	+ Add New
Susan	Foreman	Roadwork	
Sohn	Worker	Grading Complete	
Andrew	Operator	Grading Complete	
Tomás	Foreman	Grading Complete	
Peter	Worker		

Home Projects Map Equipment Employees Logout

Grading Complete

5 employees

6 equipment

Plum App Assigned

Home Projects Map Equipment Employees Logout

Excavators

Dump Trucks

AS-SI

Home Projects Map Equipment Employees Logout

Projects → Grading Complete

Equipment	Employees	Materials
OW-100	Sohn	5 Loads Mail
AS-SI	Alex	2 Trucks Cement
	Tomás	3 Loads Gravel

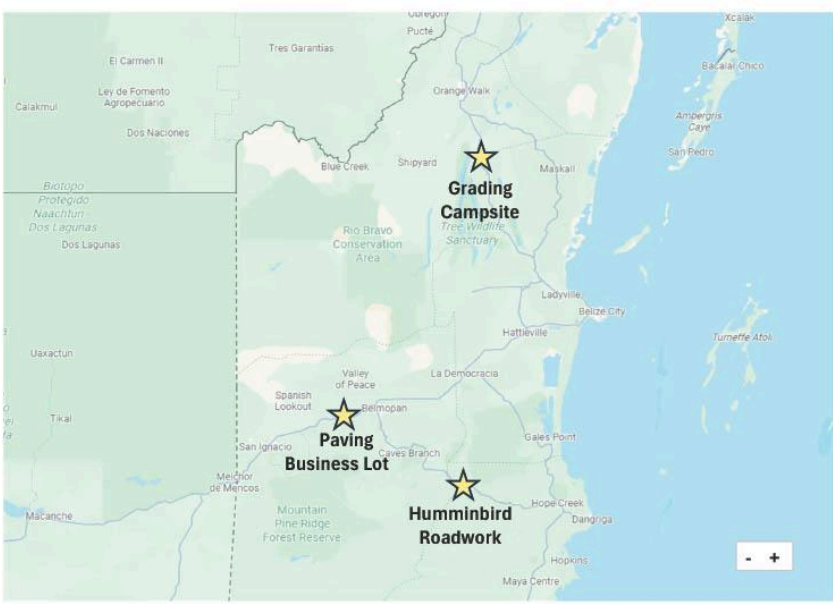
52% Complete

Progress Logs

Sep 20 Delay

Lo-Fi Prototype

Home	Add New Map Dashboard	
Projects	★ Grading Campsite  EX-2  DT-1  John  Andrew  Tomas  Maul - 1 +  Cement - 2 +  Stone - 3 + ★ Hummingbird Roadwork  CT-1  G-1  Juan  Matt  Bricks - 2 +  Cement - 2 + ★ Paving Business Lot  DT-3  CT-2	Unassigned  EX-1  EX-3  DT-2  Peter  Brad  Alan  Maul  Stone  Cement  Bricks
Logout		

Home	Map Dashboard
Projects	
Equipment	
Employees	
Material List	
Logout	

Home	Equipment > EX-1	Back																
Projects	EX-1																	
Equipment	 Model Number: 14DF56 Brand: CAT Purchase Date: 1-Apr-22 Status: Working Condition	Edit																
Employees	Maintenance Log																	
Material List	<table border="1"> <thead> <tr> <th>Date</th> <th>Issue</th> <th>Urgency</th> <th></th> </tr> </thead> <tbody> <tr> <td>4-Jan-24</td> <td>Oil Leak</td> <td>Severe</td> <td> </td> </tr> <tr> <td>8-Sep-24</td> <td>Broken AC</td> <td>Minor</td> <td> </td> </tr> <tr> <td>21-Apr-23</td> <td>Transmission Shifting</td> <td>Fixed</td> <td> </td> </tr> </tbody> </table>		Date	Issue	Urgency		4-Jan-24	Oil Leak	Severe	 	8-Sep-24	Broken AC	Minor	 	21-Apr-23	Transmission Shifting	Fixed	 
Date	Issue	Urgency																
4-Jan-24	Oil Leak	Severe	 															
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21-Apr-23	Transmission Shifting	Fixed	 															
Logout																		

Home	Add New		
Projects			
Equipment			
Employees			
Material List			
Logout			

Medium-Fi Prototype



Dashboard

Projects

Equipment

Employees

Material List

Recent Projects

Road Paving - Hummingbird Hwy, 77% Complete

Building Construction - Coastal Road, 62% Complete

Road Work - Red Bank, 23% Complete



Employees

Projects

Equipment

Employees

Material List

Name	Position	Actions
Franz smith	Foreman	Edit
Funny Valentine	Machine Operator	Edit
Rank Davis	Brick Layer	Edit
Larry Fuller	Plumbing	Edit

Add Employee



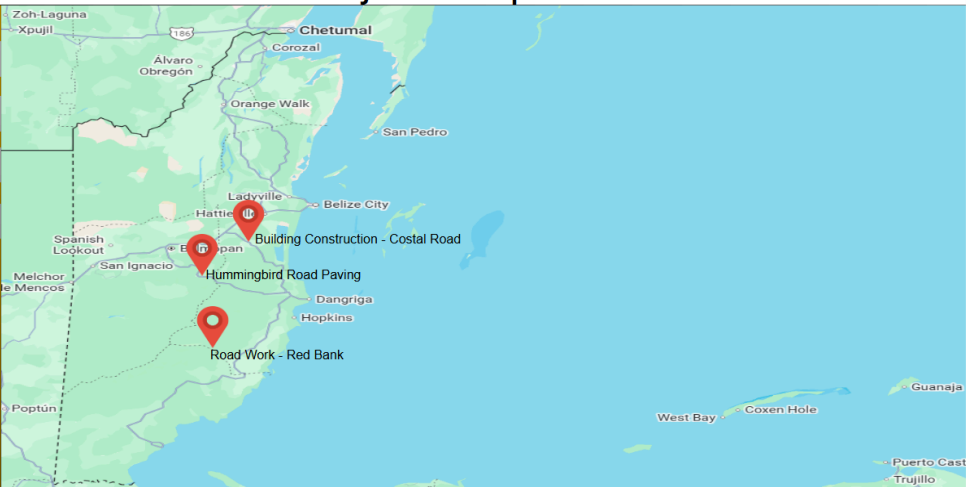
Projects Map View

Projects

Equipment

Employees

Material List



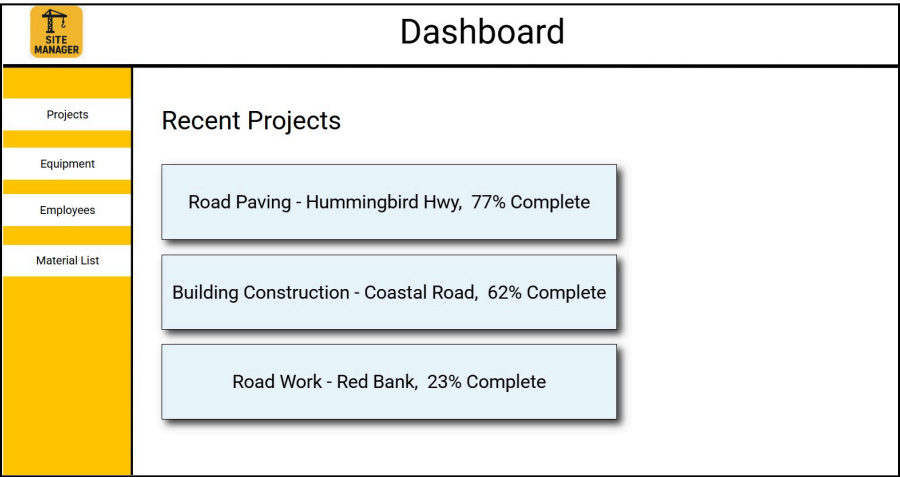
Back

Major Usability Problems Addressed

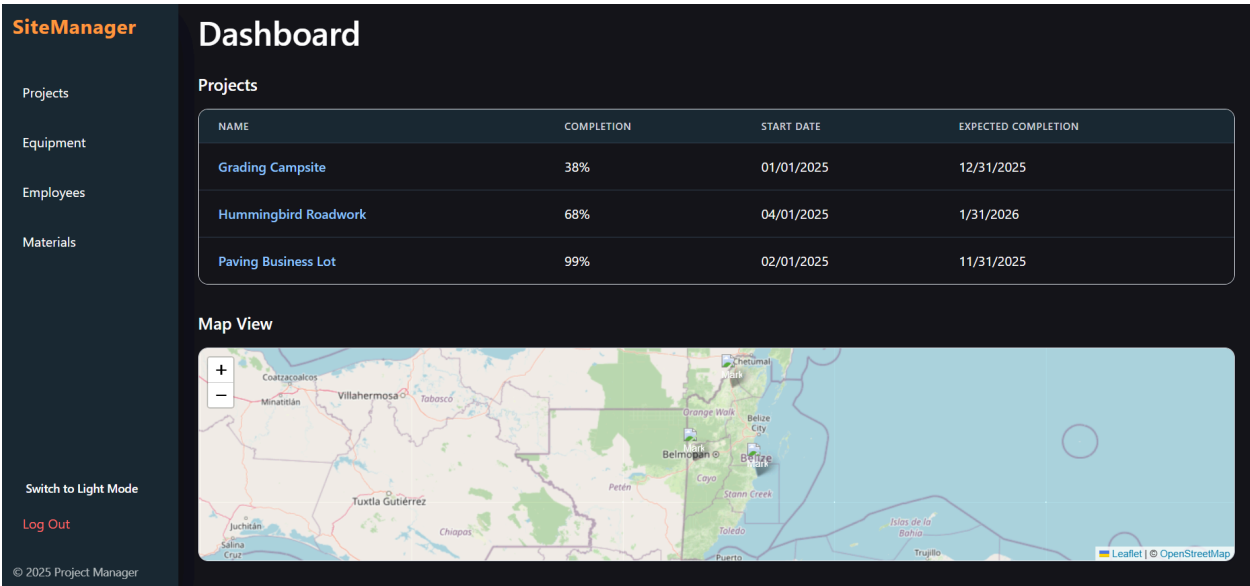
Map View

Heuristic: Match Between System and the Real World (H2) **Severity:** Major

Problem: Users struggled to visualize project locations geographically, forcing them to manually cross-reference text-based lists with external maps. This increased cognitive load and reduced efficiency, especially for managers overseeing multiple projects across different regions.



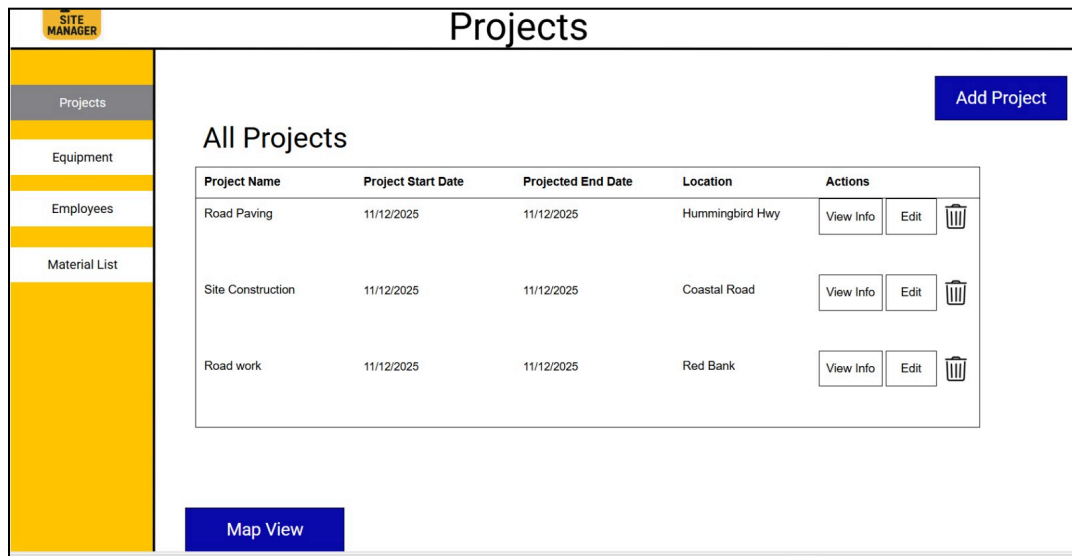
Solution: Introduced an interactive map to display project locations, enabling quick spatial assessment and improving situational awareness.



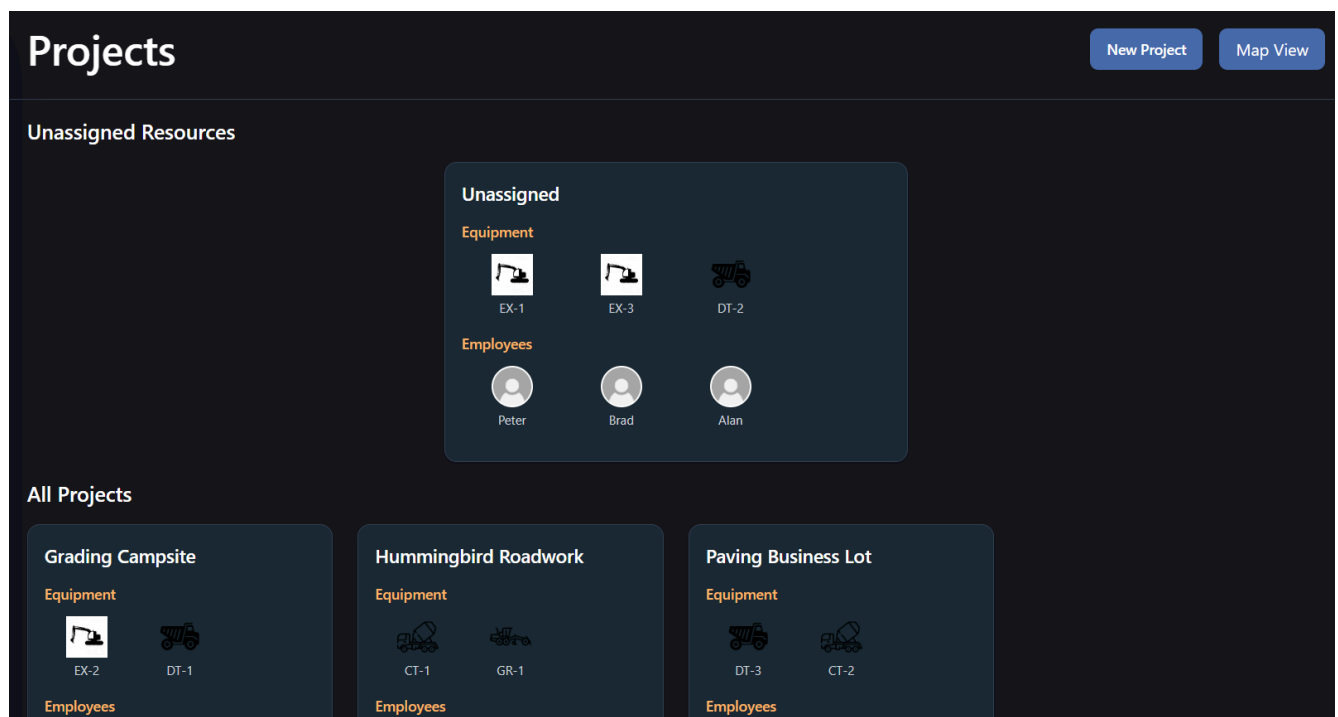
Unassigned Resources Card

Heuristic: Error Prevention (H5) & Recognition Rather Than Recall (H6) **Severity:** Major

Problem: Users lacked a centralized view of unassigned resources, leading to inefficiencies, double-booking, or overlooked equipment and employees. This increased the risk of errors in resource allocation.

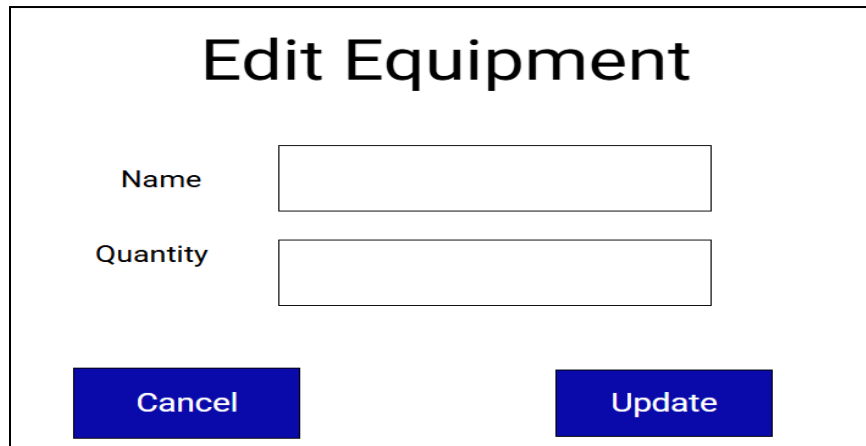


Solution: Added an "Unassigned Resources" card to provide a clear, at-a-glance view of available resources, streamlining allocation and reducing oversight.



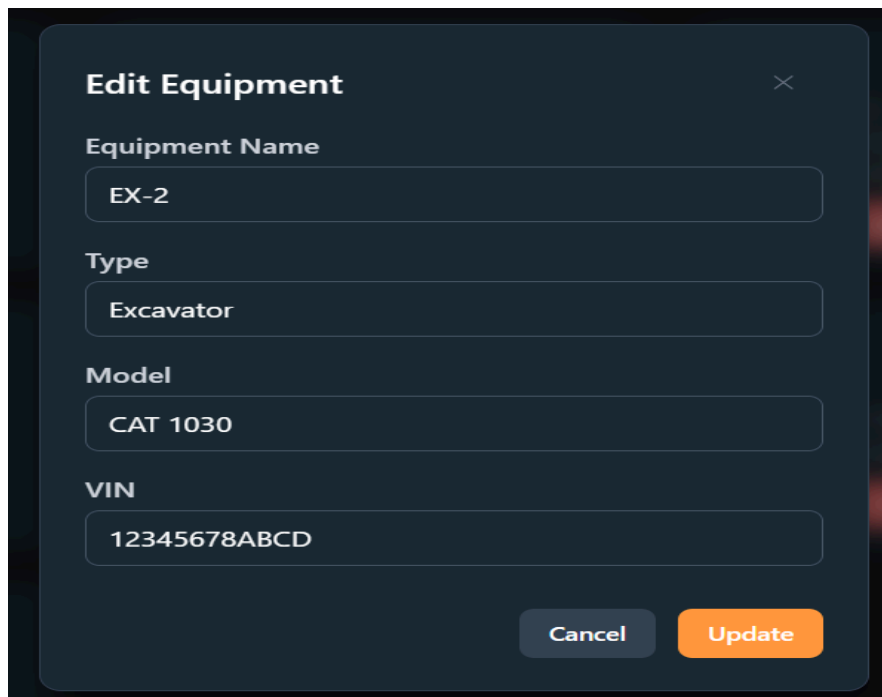
Edit Equipment

Problem: The "Edit Equipment" section only allowed basic edits (Name, Quantity), missing critical details like Type, Model, and VIN. This limited effective equipment tracking and management.



The image shows a web form titled "Edit Equipment". It contains two input fields: "Name" and "Quantity". Below the fields are two buttons: "Cancel" and "Update".

Solution: Expanded the section to include detailed fields (Type, Model, VIN), improving accuracy and usability for equipment management.



The image shows a more complex "Edit Equipment" form. It has a title bar with a close button (X). The form contains four input fields: "Equipment Name" (with value "EX-2"), "Type" (with value "Excavator"), "Model" (with value "CAT 1030"), and "VIN" (with value "12345678ABCD"). At the bottom are "Cancel" and "Update" buttons.

Detailed Project Pages

Heuristic: Visibility of System Status (H1) & Recognition Rather Than Recall (H6) **Severity:** Major

Problem: Project pages lacked structured information about progress, resources, and logs, making it difficult for users to assess project status or history. This led to miscommunication and inefficiencies.

The screenshot shows a web application interface for 'Project Info'. On the left is a yellow sidebar with a 'SITE MANAGER' logo and a list of navigation items: Projects, Equipment, Employees, and Material List. The main content area has a white background and is titled 'Project Info'. It features a blue 'Back' button at the top left. To the right, there is an 'Edit Project' section with four form fields: Project Name, Project Start Date, Projected End Date, and Location. Below these fields is a blue 'Update' button.

Solution: Added comprehensive sections (Details, Equipment, Employees, Materials, Progress Logs) to provide clarity, transparency, and easy access to project information.

The screenshot shows a more detailed project page titled 'Grading Campsite'. The sidebar is dark blue with the 'SiteManager' logo and navigation links: Projects, Equipment, Employees, and Materials. The main content area is also dark blue and contains several sections:

- Details:** Shows a progress bar at 60%, Start Date: 01/01/2025, Expected Completion: 12/31/2025, and Location: X: 17.15, Y: -89.07.
- Equipment:** Displays icons for EX-2 and DT-1.
- Employees:** Shows avatars for John, Andrew, and Tomas.
- Materials:** Lists materials and their counts: maul (3), cement (2), stone (3), and bricks (0).
- Progress Logs:** Includes an 'Add Log' button and two log entries:
 - 09/20/2025: Delay - cement mixer didn't reach on time. (Edit, Delete)
 - 06/13/2025: Finished laying foundational white maul. (Edit, Delete)

At the bottom left of the sidebar, there are links for 'Switch to Light Mode', 'Log Out', and a copyright notice '© 2025 Project Manager'.

Prototyping Tools:

- GitHub- Asynchronous cooperation is enabled by the use of a version control system. made it possible for the team to coordinate changes, share code, and collaborate effectively without having to be physically there.
- React- chosen due to its component-based architecture. immensely beneficial for designing modular, maintainable user interfaces. It enabled the rapid reuse of UI components (such as buttons and cards), which accelerated development. made sure that all application views followed standards and had a consistent visual style.
- JavaScript-It was picked due to its everyday nature. allowed the entire stack to utilize a single, standardized language (for both the React frontend and potential backend). Debugging became more manageable, and the development team's context switching decreased. enables an easy interaction with common web libraries and APIs.

Conclusion

The development of the SiteManager application was a tremendous learning experience throughout the whole Human-Computer Interaction (HCI) design process. From early manual sketches and low-fidelity planning to interactive Canva prototypes and, finally, a genuine, largely functional high-fidelity prototype, we are grateful to have learnt and mastered a wide range of skills. This project has helped us understand the need of continuous iteration, various types of testing and prototyping, and the critical impact that technological decisions have in achieving usability goals.

Our technical stack was carefully selected to promote collaboration and modularity. We utilized GitHub extensively for version control; its strong asynchronous cooperation features were really useful, allowing the team to share code, coordinate changes, and collaborate effectively without having to be physically present. React was chosen because of its component-based architecture, which is essential for developing modular and maintainable user interfaces. This architecture enabled the rapid reuse of standardized user interface elements (such as buttons, forms, and cards), resulting in significantly faster development while ensuring that all application views adhered to identical visual standards. Finally, JavaScript was selected as the critical bridge since it is widely used and allows the complete stack to communicate in a single, standardized language. This approach enabled easy interaction with common web libraries and APIs to complete the final prototype, accelerated development, simplified debugging, and reduced the team's context switching.

Even though our heuristic review and task-based storyboards validated the essential functionality, the project requires more features (such as strong error handling and full implementation of the map editing feature) before it can be considered production-ready. We greatly appreciate the extensive design and technical experience gained via this process, and we look forward to completing the remaining stages of this project for our next class.